

Indian swine flu vaccine

The Department of Health Research is actively developing an indigenous swine flu vaccine

Bioplastic race

One of the major efforts by biotech firms around the world is to find a cheap, bio-degradable substitute for plastic

Green BT

The Green Revolution was powered by chemicals in the form of fertilizers and pesticides. The environmental cost of this reliance on chemicals is only now being appreciated.

• Biofertilizers

Rhizobium bacteria absorb atmospheric nitrogen and makes it available in a form that can be absorbed by plants. Nitrogen is an essential nutrient for plants, and a major component of commercial fertilizers.

Rhizobium naturally attaches itself to legumes by forming root nodules, but through BT it is possible to increase the number of such nodules, thus increasing the rate of nitrogen absorption. Similarly, non-leguminous plants are associated with Azetobacter bacteria for nitrogen absorption. All that is needed is for seeds to be coated with these bacteria prior to sowing.

• Biopesticides

Bacillus thuringiensis is a bacterium that is used as a natural pesticide because it infects many pests, and can be sprayed over crops instead of regular pesticides. Going a step further is Bollgard BT cotton. This is a product of Monsanto Company, and the first GM crop to be planted in India. The Bt in "Bt cotton" stands for Bacillus thuringiensis, which is the source of the gene inserted into the cotton DNA, that causes the production of a protein that is toxic to the pest American Bollworm that destroys cotton balls. This innate resistance reduces the need for pesticides.

• GM livestock

Besides plants and MOs, BT has also modified large animals, mainly farm animals such as cows, pigs, goats, etc. High yielding cattle are in demand for selective breeding among dairy farmers; this is an ancient practice which takes a long time to produce results. BT speeds things up by genetic manipulation. Besides improved milk production, the nature of the milk can also be modified.

Cows have been genetically modified to carry human genes responsible for the creation of lactoferrin – an iron carrying protein that is essential for infants. The milk produced by these cows is iron-rich in contrast to normal cow milk, making it ideal infant food (to substitute human milk).

Spider silk is acknowledged to be the strongest natural fibre, and this has many potential applications where a right mix of strength and weight is needed – bullet proof vests come to mind first. By genetically modifying goats with spider genes, spider silk is produced along with goat's milk in much larger quantities than can be produced by spiders. Needless to say, the issue of animal rights is a major stumbling block in this area.

Red BT

The decoding of the human genome has allowed scientists to identify the genetic basis of human susceptibility to many diseases, but the full benefits of these findings are still some time away. Even so, BT has already contributed to healthcare in many ways:

• Testing technologies

The traditional method of testing for disease involves taking a sample (blood, sputum, etc.) and culturing it over a period of time to allow microbial growth, and then identifying the organisms.

This delay has been dramatically reduced by using tests that directly check for unique patterns of proteins or DNA (much like anti-virus software flags suspicious programs). Monoclonal antibody testing is based on the principle of the unique antibody-antigen relation. Polymerase Chain Reaction (PCR) is another BT product that is used to rapidly make copies of genetic material

allowing even minute samples to be used for testing. It is the process behind such techniques

as DNA fingerprinting which can be used to uniquely identify any living organism.

PCR can also be used to test genetic material for identifying disease agents – the test to detect the Swine flu virus (H1N1) is based on PCR.

The use of animals as test subjects is controversial, and yet the only way to ensure a drug is fit for human trial. The search for a cancer treatment has benefited from a special type of mouse called Oncomouse or Harvard Mouse, which is a BT product. This GM mouse has a human gene (oncogene) that makes it susceptible to cancers, and hence is an ideal test subject for anti-cancer research.

• Medicines

BT has made it possible to generate medicines for human consumption by genetically modifying MOs, and insulin generating MOs have already been mentioned. Other medicines that are BT products include recombinant EPO (Erythropoietin) a protein that is responsible for red blood cell generation. EPO is created on a large scale by inserting the human gene responsible for EPO production to Chinese hamster ovary cells which are then cultured. EPO is the most widely sold BT medicine, and is needed for anaemic patients – such as those suffering from kidney failure. We have mentioned the potential uses of bioplastics. In the medical arena, bioplastic surgical thread is used for stitching up surgical wounds. It gradually self dissolves, saving the patient another hospital visit.

Blue BT

Marine organisms have been a source of food and medicines since ancient times. Different forms of algae serve as protein rich food, a source of gelling agents used as additive in ice creams, as lubricants, etc. The wonder properties of Spirulina

were the topic of much discussion in the recent past. "The most powerful food in the world." Spirulina is an algae that is rich in many essential nutrients, proteins, vitamins, iron, etc. Medicines from marine life include a painkiller that is actually a neurotoxin and part of the defence mechanism of cone snails.

Algae are being seen as an important source of biofuels, especially since algae seem to thrive on waste water – irrespective of the source: either from factories, mines or residential complexes.

The Future

The future will see greener alternatives replace most, if not all, environmentally unfriendly chemicals and fuels, thanks to BT. Butanol, which is seen as a replacement for petrol, and ethanol are expected to be produced in larger quantities using GM bacteria.

It needs to be noted that unlike most other new products, BT products have a very long gestation period from lab to market, and even then public opinion can drown down scientific reasoning – particularly where food is concerned. So while many interesting concepts are discovered, applied, tried and tested in the lab, they need not survive the trip to the market shelf. Some BT food products in the pipeline: nutrition enhanced food: potatoes containing vaccines against hepatitis virus, rice fortified with vitamins, iron and omega3 fatty acids, etc., drought resistant GM corn – a result of inserting a bacterial gene that imparts resistance to the organism from adverse environments into the corn DNA.

And some wishful thinking doesn't seem too far-fetched. So far, human genes have been transferred to MOs to enable them to produce human products. In the future, one can expect the reverse – genes coding for the best features of plants and animals transferred into the human, creating a man that can run as fast as a cheetah and fly as well as a bird. **d**

Enzymes create the "stone washed" look in jeans, originally done by pumice stones

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